

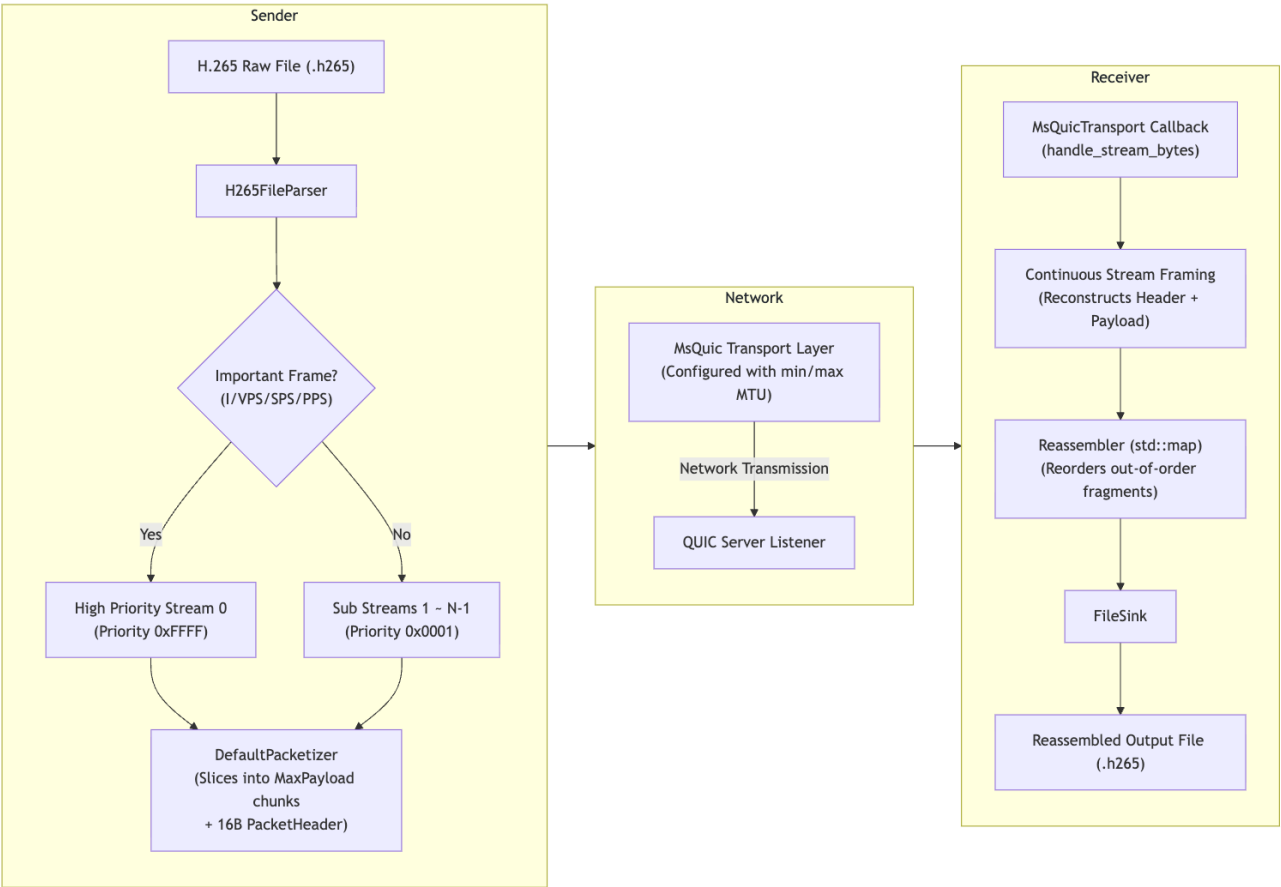
QUICMediaTransmissionSystem

This is a C++ media streaming prototype designed to transmit raw H.265/HEVC video streams over the QUIC protocol. Built on Microsoft's **MsQuic** library, the project implements custom application-layer fragmentation, multi-stream prioritization, MTU control, and tracking for networking experiments.

System Architecture

The framework is decoupled into a modular architecture:

- **Media Parser (`H265FileParser`)**: Scans input files for Annex-B start codes (`00 00 01` or `00 00 00 01`) and extracts individual NALUs.
- **Application Protocol Framing (`PacketHeader` / `DefaultPacketizer`)**: Slices large NALUs into MTU-compliant fragments prepended with a 16-byte aligned packet header.
- **Priority Scheduler**: Routes critical frames (I-frames, VPS, SPS, PPS) to the high-priority QUIC Stream 0, while spreading standard P/B-frames across sub-streams.
- **Transport Layer (`MsQuicTransport`)**: Manages the life cycle of MsQuic connections, streams, parameters (such as MTU constraints), and asynchronous I/O callbacks.
- **Receiver & Reassembler (`Reassembler` / `FileSink`)**: Buffers fragments into a non-blocking map, enforces in-order reassembly of NALUs, and writes finalized streams to output.



Environment Setup & Installation

1. Prerequisites

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- Linux OS (Ubuntu 20.04+ recommended)
- CMake 3.16+
- Compiled MsQuic and server TLS credentials (e.g. .crt, .key files)

See [docs/INSTALL.md](#) for detailed installation and build instructions for MsQuic and the media system.

2. Compilation

If the environment and credentials are ready, configure and compile:

```
# From project root directory
cmake -S . -B build \
  -DCMAKE_BUILD_TYPE=Debug \
  -DQUIC_ENABLE_LOGGING=ON \
  -DQUIC_BUILD_TOOLS=ON

cmake --build build -j
```

Usage Guide

1. Run the Server

Start the server listener first to receive the streamed output:

```
./moq_server <host> <port> <output.h265> <cert_file> <key_file>
```

- **Example:**

```
./moq_server 0.0.0.0 4433 output.h265 ./server.crt ./server.key
```

2. Run the Client

Stream the input H.265 file to the server:

```
./moq_client <host> <port> <input.h265> [num_streams] [chunk_size] [min_mtu] [max_mtu]
```

- **num_streams** (*Optional*): The total number of streams to open (Default: **2**, minimum **2** to split priority flows).
- **chunk_size** (*Optional*): Payload fragment size limit in bytes (Default: **1200**).
- **min_mtu** / **max_mtu** (*Optional*): Forces MSQuic to override PMTUD and constrain the wire packet sizes.

- **Example:**

```
./moq_client localhost 4433 input.h265 4 1000 1200 1200
```